

PROCESS INTELLIGENCE ENGINE (PIE) — AI & INTELLIGENCE ARCHITECTURE

Three Controlled AI Interaction Pathways: Extraction, Evidence Retrieval (RAG), and Natural Language Query

PATHWAY 1: 9-STAGE PIPELINE

1. Input Process Description

Raw unstructured text from industrial operations

2. Validation & Normalization

Regex cleanup, strip punctuation, lowercase

3. SHA-256 Deduplication

Idempotent hash check prevents duplicate runs

4. Groq LLM Feature Extraction

JSON-mode ratings (1–5) on 8 rubric factors +
3–5 factual claims + rationale & confidence

5. Feature Normalization

Ordinals mapped deterministically to [0.0, 1.0]

6. Deterministic Rubric Scoring

Rubric YAML weights × signed direction;
Automate / Augment / Human-Led decision

7. Safety Override Enforcement

Caps at Augment if Safety=5 & Judgment≥4

8. Persistence & Audit Log

Saves score, factors, run ID, inputs hash

9. Portfolio Re-Ranking

Exact ranks (1..N) and percentiles computed

PATHWAY 2: EVIDENCE & RAG

1. Extracted Factual Claims

Claims generated by LLM during Stage 4

2. Dual-Engine Retrieval

Lexical keyword matching +
pgvector cosine similarity (all-mpnet-base-v2)

3. 25 Industrial Corpus Docs

Ground-truth engineering & operational standards

4. Top-K Candidate Chunks

Relevant paragraph segments (500 tokens)

5. LLM Verbatim Quote Selection

Prompted to return exact quote or NO_QUOTE

6. Deterministic Quote Verification

Exact substring check: normalized(quote)
in normalized(chunk_text)

7. Provenance Record Creation

Link verified quote -> chunk -> source URL/year

8. Audit Trust Chain

Claims flagged supported=True/False with full citation

PATHWAY 3: NATURAL LANGUAGE QUERY

1. User Business Question

"Which maintenance processes should we automate?"

2. Closed Intent Classification

Groq classifies into 7 whitelisted intents:
rank_top, filter_by_band, explain_process,
portfolio_stats, compare, open_research, unmappable

3. Strongly Typed QueryPlan

Pydantic model with validated parameters
(limit, band filter, metric, target_id)

4. Whitelisted Query Executor

NO TEXT-TO-SQL. Pre-compiled SQLAlchemy
parameterized queries mapped to intent.

5. Secure Database Execution

Direct, parameterized PostgreSQL execution

6. LLM Prose Explanation

LLM explains pre-computed numbers;
Zero calculation done by LLM.

7. UI Response Rendering

Tabular data + prose + debug QueryPlan